

NIMCET COMPUTER SCIENCE – COMPLETE NOTES, FORMULAS, PYQs & ANALYSIS

Exam Context: NIMCET 2025 was conducted by **NIT Trichy**, and they introduced **deeper conceptual questions on Computer Organization & Architecture**. Expect this trend to continue in 2026.

TOPIC-WISE IMPORTANCE ANALYSIS (Based on PYQ Trends)

Topic	Approx. Questions	Trend (2025)	Priority
Computer Organization & Architecture	4-6	▲ Increased & Deeper	★★★★★★
Data Structures	4-5	Stable	★★★★★★
C Programming & Output	3-5	Stable	★★★★★★
Operating Systems	3-4	Stable	★★★★★
DBMS (SQL + Relational Algebra)	2-3	Stable	★★★★★
Computer Networks	2-3	Stable	★★★★★
Number System & Boolean Algebra	2-3	Stable	★★★★★
Software Engineering	1-2	Low	★★★
Theory of Computation / Compiler	0-1	Rare	★★

COMPUTER ORGANIZATION & ARCHITECTURE (HIGH PRIORITY – 2025 TREND)



SHORT NOTES

1.1 Number Representation in Computer

Fixed Point Representation:

- **Sign-Magnitude:** MSB = sign bit (0 = +ve, 1 = -ve), rest = magnitude
- **1's Complement:** Invert all bits for negative
- **2's Complement:** 1's complement + 1 (most widely used)

Ranges for n-bit representation:

Method	Range
Unsigned	0 to $2^n - 1$
Sign-Magnitude	$-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$
1's Complement	$-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$
2's Complement	-2^{n-1} to $+(2^{n-1} - 1)$

Note: 2's complement has **one extra negative number** and **no negative zero**.

Floating Point (IEEE 754):

Format	Total Bits	Sign	Exponent	Mantissa	Bias
Single Precision	32	1	8	23	127
Double Precision	64	1	11	52	1023

$$\text{Value} = (-1)^S \times 1.M \times 2^{(E - \text{Bias})}$$

Special Values:

- E = all 0s, M = 0 → **Zero**
- E = all 0s, M ≠ 0 → **Denormalized number**
- E = all 1s, M = 0 → **±Infinity**
- E = all 1s, M ≠ 0 → **NaN**

1.2 Booth's Algorithm (Multiplication)

Used for **signed binary multiplication** using 2's complement.

Steps:

1. Initialize: $A = 0$, Q = Multiplier, $Q_{-1} = 0$, M = Multiplicand, Count = n
2. Check Q_0 and Q_{-1} :
 - $Q_0Q_{-1} = 10 \rightarrow A = A - M$
 - $Q_0Q_{-1} = 01 \rightarrow A = A + M$
 - $Q_0Q_{-1} = 00 \text{ or } 11 \rightarrow$ No operation
3. Arithmetic Right Shift (A , Q , Q_{-1})
4. Decrement count. Repeat until count = 0.

1.3 CPU Organization

Types of CPU Organization:

Type	Operands in Instruction	Example
Single Accumulator	Implicit (ACC)	$\text{ADD } X \rightarrow \text{ACC} = \text{ACC} + M[X]$
General Register	Registers	$\text{ADD } R1, R2, R3$
Stack	TOS (implicit)	PUSH, POP, ADD

Instruction Format Types:

- **3-Address:** $\text{ADD } R1, R2, R3 \rightarrow R1 = R2 + R3$
- **2-Address:** $\text{ADD } R1, R2 \rightarrow R1 = R1 + R2$
- **1-Address:** $\text{ADD } X \rightarrow \text{ACC} = \text{ACC} + M[X]$
- **0-Address:** Stack-based $\rightarrow \text{PUSH } A, \text{ADD}$ (operates on TOS)

To evaluate $X = (A + B) \times (C + D)$:

Format	No. of Instructions
3-Address	3
2-Address	4-5
1-Address	6-7
0-Address	7-8

1.4 Addressing Modes

Mode	Effective Address	Description
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Mode	Effective Address	Description
Immediate	Operand = value itself	No memory access for operand
Direct	EA = Address field	One memory access
Indirect	EA = M[Address field]	Two memory accesses
Register	EA = Register	Fast, no memory access
Register Indirect	EA = Contents of Register	One memory access
Displacement (Indexed/Base/Relative)	EA = A + Register content	Used in arrays, relocation
Auto-increment	EA = (R), then R = R + 1	Sequential access
Auto-decrement	R = R - 1, then EA = (R)	Stack implementation
PC-Relative	EA = PC + Offset	Used in branch instructions

1.5 Instruction Cycle

Fetch → Decode → Execute → Memory Access → Write Back

With Interrupt:

Fetch → Decode → Execute → [Check Interrupt]

Micro-operations for Fetch:

T0: MAR ← PC

T1: MBR ← M[MAR], PC ← PC + 1

T2: IR ← MBR

1.6 Pipelining

Key Formulas:

Non-pipelined time for n instructions = $n \times k \times t_p$

Where k = number of stages, t_p = time per stage (clock cycle)

Pipelined time = $(k + n - 1) \times t_p$

Speedup S = $(n \times k \times t_p) / ((k + n - 1) \times t_p)$

For large n: $S \rightarrow k$ (ideal speedup = number of stages)

Throughput = $n / ((k + n - 1) \times t_p)$

$$\text{Efficiency} = \text{Speedup} / k = n / (k + n - 1)$$

If stages have unequal delays:

$$\text{Clock cycle} = \max(\text{stage delays}) + \text{Buffer delay}$$

Pipeline Hazards:

1. **Structural Hazard** – Resource conflict
2. **Data Hazard** – Data dependency (RAW, WAR, WAW)
 - RAW (Read After Write) – True dependency – **most common**
 - WAR (Write After Read) – Anti-dependency
 - WAW (Write After Write) – Output dependency
3. **Control Hazard** – Branch instructions

Solutions:

- **Forwarding/Bypassing** – For data hazards
- **Stalling (Bubble)** – Insert NOP
- **Branch Prediction** – For control hazards
- **Delayed Branch** – Reorder instructions

Stall Cycles:

$$\text{Penalty per branch} = (\text{Misprediction rate}) \times (\text{Branch penalty cycles})$$

1.7 Memory Hierarchy

Registers → Cache → Main Memory (RAM) → Secondary Storage (Disk)

← Faster, Smaller, Costlier →

← Slower, Larger, Cheaper →

1.8 Cache Memory

Mapping Techniques:

Technique	Tag Bits	Index Bits	Block Offset
Direct Mapped	Address – (Index + Offset)	$\log_2(\text{No. of lines})$	$\log_2(\text{Block size})$
Fully Associative	Address – Offset	None	$\log_2(\text{Block size})$
Set Associative (k-way)	Address – (Set index + Offset)	$\log_2(\text{No. of sets})$	$\log_2(\text{Block size})$

Key Formulas:

Number of cache lines = Cache size / Block size

For Direct Mapped:

Number of lines = Cache size / Block size

For k-way Set Associative:

Number of sets = (Cache size) / (Block size × k)

Tag bits = Total address bits - Index bits - Offset bits

Block offset bits = $\log_2$ (Block size in bytes)

Word offset bits = $\log_2$ (Words per block)

Hit Rate and Miss Rate:

Hit Rate (h) + Miss Rate (m) = 1

$$\begin{aligned}\text{Effective Access Time (EAT)} &= h \times t_c + (1 - h) \times (t_c + t_m) \\ &= t_c + (1 - h) \times t_m\end{aligned}$$

Where t_c = cache access time, t_m = memory access time (miss penalty)

For hierarchical:

$$\text{EAT} = h_1 \times t_1 + (1 - h_1) \times h_2 \times t_2 + (1 - h_1)(1 - h_2) \times t_3$$

AMAT (Average Memory Access Time):

AMAT = Hit time + Miss rate × Miss penalty

Replacement Policies: LRU, FIFO, LFU, Random

Write Policies:

- **Write-through:** Write to both cache and memory simultaneously
- **Write-back:** Write only to cache; memory updated on eviction
- **Write-allocate:** On miss, load block to cache then write
- **No-write-allocate:** On miss, write directly to memory

1.9 Virtual Memory

Virtual Address → Page Number + Page Offset

Physical Address → Frame Number + Page Offset

Page Table Size = (Number of pages) × (Page Table Entry size)

Number of pages = $2^{(\text{virtual address bits})} / \text{Page size}$

Number of frames = $2^{(\text{physical address bits})} / \text{Page size}$

TLB (Translation Lookaside Buffer):

$$\text{EAT} = \text{TLB_hit_rate} \times (\text{TLB_time} + \text{Memory_time}) + \text{TLB_miss_rate} \times (\text{TLB_time} + 2 \times \text{Memory_time})$$

Page Fault Rate:

$$\text{EAT} = (1 - p) \times \text{Memory_access_time} + p \times \text{Page_fault_service_time}$$

1.10 I/O Organization

Methods:

1. **Programmed I/O** – CPU busy waits (polling)
2. **Interrupt-driven I/O** – CPU interrupted when I/O ready
3. **DMA (Direct Memory Access)** – Data transfer without CPU

DMA Modes:

- **Burst mode** – Entire block transferred, CPU halted
- **Cycle stealing** – One word per bus grant
- **Transparent/Interleaved** – Uses bus when CPU doesn't need it

I/O Mapping:

- **Memory-mapped I/O** – Same address space, same instructions
 - **Isolated (Port-mapped) I/O** – Separate address space, special instructions (IN/OUT)
-

1.11 Instruction Set Architecture

RISC vs CISC:

Feature	RISC	CISC
Instructions	Simple, fixed-length	Complex, variable-length
Clock cycles/instruction	Mostly 1	Multiple
Addressing modes	Few	Many
Registers	More	Fewer

Feature	RISC	CISC
Pipelining	Easy	Difficult
Examples	ARM, MIPS, SPARC	x86, VAX

1.12 Interconnection Structures (Bus)

Types:

- **Data Bus** – Carries data (bidirectional)
- **Address Bus** – Carries address (unidirectional)
- **Control Bus** – Control signals

If address bus = n bits $\rightarrow$ Addressable memory = 2^n locations

If data bus = m bits $\rightarrow$ m bits transferred at a time

Bus Arbitration:

- Daisy chaining (centralized)
 - Polling (centralized)
 - Independent request (distributed)
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1.13 ALU & Arithmetic Operations

Addition/Subtraction Circuits:

- **Half Adder:** Sum = $A \oplus B$, Carry = $A \cdot B$
- **Full Adder:** Sum = $A \oplus B \oplus C_{in}$, Cout = $AB + BC_{in} + AC_{in}$

Carry Lookahead Adder (CLA):

Generate: $G_i = A_i \cdot B_i$

Propagate: $P_i = A_i \oplus B_i$

Carry: $C_{i+1} = G_i + P_i \cdot C_i$

Sum: $S_i = P_i \oplus C_i$

Time for n-bit CLA:

Time = $O(\log n)$ vs Ripple Carry Adder = $O(n)$

1.14 Control Unit Design

Hardwired Control:

- Fast, complex to design, difficult to modify
- Combinational logic circuits

Microprogrammed Control:

- Slower, flexible, easy to modify
 - Control memory stores microinstructions
 - **Horizontal:** Wide, parallel, each bit controls a signal
 - **Vertical:** Encoded, narrower, needs decoder
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ESSENTIAL FORMULAS SUMMARY – COMPUTER ORGANIZATION

1. 2's complement range: $-2^{(n-1)}$ to $2^{(n-1)} - 1$
 2. IEEE 754 Single: $(-1)^S \times 1.M \times 2^{(E-127)}$
 3. Pipeline Speedup: $S = nk / (k + n - 1)$
 4. Pipeline time: $(k + n - 1) \times t_p$
 5. Max Speedup ($n \rightarrow \infty$): $S = k$
 6. AMAT = Hit_time + Miss_rate $\times$ Miss_penalty
 7. EAT = $h \times t_c + (1-h)(t_c + t_m)$
 8. Cache lines = Cache_size / Block_size
 9. Sets (k-way) = Cache_size / (Block_size $\times$ k)
 10. Page Table Size = Num_pages $\times$ Entry_size
 11. CPI_pipeline = Ideal CPI + Stall cycles per instruction
 12. Bus Bandwidth = Bus width $\times$ Bus frequency
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PYQs – COMPUTER ORGANIZATION

- Q1.** In IEEE 754 single precision floating point representation, how many bits are used for mantissa (significand)?
- Q2.** In a pipelined processor with 5 stages, if each stage takes 2 ns, what is the time to execute 100 instructions?
- Q3.** Which addressing mode is used for relocating programs in memory without changing the code?
- Q4.** What is the speedup of a 6-stage pipeline processing 100 instructions compared to non-pipelined execution?

- Q05.** In a 2-way set associative cache, if cache size is 64 KB and block size is 32 bytes, how many sets are there?
- Q06.** In 8-bit 2's complement representation, the range of numbers that can be represented is?
- Q07.** Which data hazard is also called true data dependency?
- Q08.** In a direct-mapped cache with 256 lines and block size of 4 words (16 bytes), how many tag bits are needed for a 32-bit address?
- Q09.** DMA is used to transfer data between _____ and _____ without CPU intervention.
- Q10.** What is the effective memory access time if cache hit ratio is 0.9, cache access time is 10 ns, and main memory access time is 100 ns?
- Q11.** In Booth's multiplication algorithm, when $Q_0Q_{-1} = 10$, what operation is performed?
- Q12.** Which of the following is a characteristic of RISC architecture?
(a) Variable length instructions (b) More addressing modes (c) Hardwired control unit (d) Microprogrammed control
- Q13.** What is the purpose of a TLB in virtual memory systems?
- Q14.** The number of page table entries in a system with 32-bit virtual address and 4 KB page size is?
- Q15.** What is the difference between write-through and write-back cache policies?
- Q16 (2025 Type - Conceptual).** In a 4-stage pipeline, if a branch instruction is detected at the decode stage and the branch penalty is 2 cycles, how many stall cycles are introduced for 1 branch instruction?
- Q17 (2025 Type).** Explain the difference between hardwired and microprogrammed control units. Which is faster?
- Q18 (2025 Type).** In a system with 20-bit virtual address and 16-bit physical address, page size 1 KB, what is the maximum number of frames?
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DATA STRUCTURES

SHORT NOTES

2.1 Arrays

1D Array:

Address of $A[i] = \text{Base} + (i - \text{lower_bound}) \times \text{element_size}$

2D Array (Row Major):

Address of $A[i][j] = \text{Base} + [(i - \text{lr}) \times \text{Nc} + (j - \text{lc})] \times \text{size}$

Where $\text{Nc} = \text{number of columns} = \text{uc} - \text{lc} + 1$

2D Array (Column Major):

Address of $A[i][j] = \text{Base} + [(j - \text{lc}) \times \text{Nr} + (i - \text{lr})] \times \text{size}$

Where $\text{Nr} = \text{number of rows} = \text{ur} - \text{lr} + 1$

Total elements in 2D array: $\text{Nr} \times \text{Nc} = (\text{ur} - \text{lr} + 1) \times (\text{uc} - \text{lc} + 1)$

2.2 Stacks

- **LIFO** (Last In, First Out)
- Operations: **Push, Pop, Peek/Top, isEmpty, isFull**
- **Overflow:** Push on full stack
- **Underflow:** Pop from empty stack
- Applications: Function calls, recursion, expression evaluation, undo, backtracking, parenthesis matching

Using Array: $\text{Top} = -1$ (empty), $\text{Top} = n-1$ (full)

2.3 Queues

- **FIFO** (First In, First Out)
- Operations: **Enqueue, Dequeue**
- **Types:** Simple, Circular, Priority, Deque

Circular Queue:

$\text{Rear} = (\text{Rear} + 1) \% \text{SIZE}$

$\text{Front} = (\text{Front} + 1) \% \text{SIZE}$

Full: $(\text{Rear} + 1) \% \text{SIZE} == \text{Front}$

Empty: $\text{Front} == -1$ (or $\text{Front} == \text{Rear}$ in some implementations)

Max elements = $\text{SIZE} - 1$ (one slot wasted) or use count variable

2.4 Linked List

Type	Features
Singly	Each node $\rightarrow$ Data + Next pointer
Doubly	Each node $\rightarrow$ Prev + Data + Next
Circular Singly	Last node points to first
Circular Doubly	Last $\leftrightarrow$ First

Key Operations Complexity:

Operation	Array	Linked List
Access by index	$O(1)$	$O(n)$
Search	$O(n)$ / $O(\log n)$ sorted	$O(n)$
Insert at beginning	$O(n)$	$O(1)$
Insert at end	$O(1)$ amortized	$O(n)$ SLL / $O(1)$ with tail
Delete	$O(n)$	$O(1)$ if node given

2.5 Trees

Binary Tree Properties:

1. Max nodes at level $i = 2^i$ (root at level 0)
2. Max nodes in tree of height $h = 2^{(h+1)} - 1$
3. Min height for n nodes $= \lfloor \log_2 n \rfloor$
4. For n nodes: number of NULL pointers $= n + 1$
5. Number of leaf nodes (n_0) $= n_2 + 1$
Where n_2 = nodes with 2 children
6. Total nodes $n = n_0 + n_1 + n_2$

Tree Traversals:

- **Inorder (LNR):** Left $\rightarrow$ Node $\rightarrow$ Right $\rightarrow$ gives **sorted order for BST**
- **Preorder (NLR):** Node $\rightarrow$ Left $\rightarrow$ Right
- **Postorder (LRN):** Left $\rightarrow$ Right $\rightarrow$ Node
- **Level Order:** BFS using queue

Binary Search Tree (BST):

Search, Insert, Delete:

Average: $O(\log n)$

Worst (skewed): $O(n)$

AVL Tree (Self-balancing BST):

Balance Factor = $\text{Height}(\text{left}) - \text{Height}(\text{right}) \in \{-1, 0, 1\}$

Minimum nodes for height h :

$$N(h) = N(h-1) + N(h-2) + 1$$

$$N(0) = 1, N(1) = 2$$

Maximum height for n nodes: $O(\log n)$ — specifically $1.44 \log_2(n)$

Rotations: LL, RR, LR, RL

Heap (Max-Heap / Min-Heap):

Complete binary tree

Max-Heap: Parent $\geq$ Children

Min-Heap: Parent $\leq$ Children

Array representation (1-indexed):

$$\text{Parent of } i = \lfloor i/2 \rfloor$$

$$\text{Left child of } i = 2i$$

$$\text{Right child of } i = 2i + 1$$

Insert: $O(\log n)$ — insert at end, heapify up

Delete max/min: $O(\log n)$ — replace root with last, heapify down

Build heap: $O(n)$

2.6 Graph

Representations:

- **Adjacency Matrix:** Space $O(V^2)$, edge check $O(1)$
- **Adjacency List:** Space $O(V + E)$, suitable for sparse graphs

Traversals:

- **BFS:** Queue-based, $O(V + E)$, finds shortest path in unweighted graph
- **DFS:** Stack/Recursion-based, $O(V + E)$, cycle detection, topological sort

Key Formulas:

Undirected graph:

Max edges = $V(V-1)/2$ (complete graph)

Sum of degrees = $2 \times |E|$

Directed graph:

Max edges = $V(V-1)$

Sum of in-degrees = Sum of out-degrees = $|E|$

Tree: $|E| = |V| - 1$ (connected, acyclic)

Shortest Path Algorithms:

Algorithm	Type	Time Complexity	Negative Weights
BFS	Unweighted	$O(V+E)$	N/A
Dijkstra	Single source, non-negative	$O(V^2)$ or $O((V+E)\log V)$	No
Bellman-Ford	Single source	$O(VE)$	Yes
Floyd-Warshall	All pairs	$O(V^3)$	Yes (no neg cycle)

MST Algorithms:

Algorithm	Approach	Time Complexity
Kruskal's	Edge-based (sort edges, union-find)	$O(E \log E)$
Prim's	Vertex-based (grow tree)	$O(V^2)$ or $O(E \log V)$

Topological Sort:

- Only for **DAG** (Directed Acyclic Graph)
- BFS (Kahn's): Use in-degree
- DFS: Reverse of finish order
- Time: $O(V + E)$

2.7 Sorting Algorithms

Algorithm	Best	Average	Worst	Space	Stable
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Yes
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$	No
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Yes

Algorithm	Best	Average	Worst	Space	Stable
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	Yes
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$	No
Heap Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(1)$	No
Counting Sort	$O(n+k)$	$O(n+k)$	$O(n+k)$	$O(k)$	Yes
Radix Sort	$O(d(n+k))$	$O(d(n+k))$	$O(d(n+k))$	$O(n+k)$	Yes

Lower bound for comparison-based sorting = $\Omega(n \log n)$

2.8 Searching

Linear Search: $O(n)$ – any array

Binary Search: $O(\log n)$ – sorted array only

Hashing: $O(1)$ average

Hashing:

Load Factor (α) = n / m (n = elements, m = table size)

Collision Resolution:

1. Chaining (Open Hashing): Avg search = $1 + \alpha/2$ (successful)

2. Open Addressing (Closed Hashing):

– Linear Probing: $h(k, i) = (h'(k) + i) \bmod m$ – causes primary clustering

– Quadratic Probing: $h(k, i) = (h'(k) + c_1 i + c_2 i^2) \bmod m$

– Double Hashing: $h(k, i) = (h_1(k) + i \cdot h_2(k)) \bmod m$

2.9 Expression Conversion

Precedence (high to low): $^ > * / > + -$

Infix to Postfix (using stack):

- Operand $\rightarrow$ output
- Operator $\rightarrow$ pop from stack while top has $\geq$ precedence, then push
- ($\rightarrow$ push
-) $\rightarrow$ pop until (

Postfix Evaluation (using stack):

- Operand → push
 - Operator → pop two operands, compute, push result
-

1 2
3 4

ESSENTIAL FORMULAS – DATA STRUCTURES

1. Array 2D Row Major: $\text{Base} + [(i-1)r \times Nc + (j-1c)] \times \text{size}$
2. Binary Tree: Max nodes at level $i = 2^i$
3. Binary Tree: Max nodes of height $h = 2^{(h+1)} - 1$
4. Leaf nodes = Internal nodes with 2 children + 1
5. BST operations: Average $O(\log n)$, Worst $O(n)$
6. Heap build: $O(n)$
7. Complete graph edges: $V(V-1)/2$
8. Tree edges: $V - 1$
9. Comparison sort lower bound: $\Omega(n \log n)$
10. Binary search: $O(\log_2 n)$
11. Hashing average (chaining): $O(1 + \alpha)$
12. Number of BSTs with n nodes = Catalan(n) = $C(2n, n) / (n+1)$

Catalan Number:

$$C_n = C(2n, n) / (n + 1) = (2n)! / ((n+1)! \times n!)$$

Applications:

- Number of distinct BSTs with n keys
 - Number of valid parenthesizations
 - Number of full binary trees with $n+1$ leaves
-



PYQs – DATA STRUCTURES

01. The number of possible binary trees with 4 nodes is _____.
02. What is the minimum number of nodes in an AVL tree of height 3?
03. In a max-heap with n elements, where can the minimum element reside?
04. Which data structure is used for BFS traversal of a graph?
05. Convert the infix expression $A + B * C - D / E$ to postfix.
06. What is the time complexity of building a heap from an array of n elements?
07. In a complete binary tree with 1023 nodes, the height of the tree is _____.

- Q08.** Which sorting algorithm has the best worst-case time complexity among comparison-based sorts?
- Q09.** In a circular queue of size 10, if front = 3 and rear = 7, how many elements are in the queue?
- Q10.** What is the time complexity of searching an element in a balanced BST?
- Q11.** The address of element $A[5][3]$ in a 2D array $A[1..10][1..5]$ stored in row-major order with base address 1000 and each element taking 4 bytes is?
- Q12.** Which traversal of a BST gives elements in sorted order?
- Q13.** The maximum number of edges in a directed graph with n vertices is?
- Q14.** In a hash table of size 10 with hash function $h(k) = k \bmod 10$ and linear probing, if keys 23, 33, 43 are inserted, what is the position of 43?
- Q15.** What is the worst-case time complexity of Quick Sort and when does it occur?
- Q16.** Number of leaf nodes in a complete binary tree with n internal nodes?
- Q17.** Which is the most efficient data structure for implementing priority queue?
- Q18.** Time complexity of Dijkstra's algorithm using adjacency matrix representation?
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C PROGRAMMING

SHORT NOTES

3.1 Data Types & Sizes

Type	Size (typical 32-bit)	Range
char	1 byte	-128 to 127
unsigned char	1 byte	0 to 255
int	4 bytes	-2^{31} to $2^{31}-1$
unsigned int	4 bytes	0 to $2^{32}-1$
float	4 bytes	~7 decimal digits
double	8 bytes	~15 decimal digits

Type	Size (typical 32-bit)	Range
long long	8 bytes	-2^{63} to $2^{63}-1$

sizeof: Returns size in bytes

3.2 Operators Precedence (High to Low)

() [] -> .	(Postfix)
! ~ ++ -- + - * & (type) sizeof	(Unary – Right to Left)
* / %	(Multiplicative)
+ -	(Additive)
<< >>	(Shift)
< <= > >=	(Relational)
== !=	(Equality)
&	(Bitwise AND)
^	(Bitwise XOR)
	(Bitwise OR)
&&	(Logical AND)
	(Logical OR)
?:	(Ternary – Right to Left)
= += -= etc.	(Assignment – Right to Left)
,	(Comma)

3.3 Pointers

```
int a = 10;
int *p = &a;    // p stores address of a
*p = 20;        // a is now 20 (dereference)
```

Pointer Arithmetic:

$p + 1 \rightarrow$ moves by `sizeof(data_type)` bytes
 $p++ \rightarrow$ next element
 $p - q \rightarrow$ number of elements between them

Array and Pointers:

```
arr[i] ≡ *(arr + i)
&arr[i] ≡ arr + i
```

Pointer to Pointer:

```
int **pp = &p;  
**pp = 30; // a = 30
```

Function Pointer:

```
int (*fp)(int, int) = &add;
```

3.4 Strings

```
// Important functions (string.h):  
strlen(s) // length (excluding '\0')  
strcpy(d, s) // copy s to d  
strcat(d, s) // concatenate s to d  
strcmp(s1, s2) // compare: 0 if equal, <0 or >0  
strrev(s) // reverse (non-standard)
```

String is array of chars terminated by '\0'

```
char s[] = "hello"; // size = 6 (including '\0')
```

3.5 Recursion

```
// Factorial  
int fact(int n) {  
    if (n <= 1) return 1;  
    return n * fact(n - 1);  
}  
  
// Fibonacci  
int fib(int n) {  
    if (n <= 1) return n;  
    return fib(n-1) + fib(n-2); // Time: O(2^n)  
}  
  
// Tower of Hanoi  
// Moves = 2^n - 1
```

Recurrence Relations:

Recurrence	Solution
$T(n) = T(n-1) + c$	$O(n)$
$T(n) = T(n-1) + n$	$O(n^2)$
$T(n) = 2T(n/2) + n$	$O(n \log n)$ – Merge Sort
$T(n) = 2T(n/2) + c$	$O(n)$
$T(n) = T(n/2) + c$	$O(\log n)$ – Binary Search
$T(n) = 2T(n-1) + c$	$O(2^n)$ – Tower of Hanoi

Master Theorem: $T(n) = aT(n/b) + O(n^d)$

If $d < \log_b(a) : O(n^{\log_b(a)})$

If $d = \log_b(a) : O(n^d \times \log n)$

If $d > \log_b(a) : O(n^d)$

3.6 Storage Classes

Class	Scope	Lifetime	Default Value	Storage
auto	Local	Function	Garbage	Stack
register	Local	Function	Garbage	CPU Register
static	Local	Program	0	Data segment
extern	Global	Program	0	Data segment

3.7 Structure & Union

```
struct Point { int x; int y; };  
// Size = sum of members (+ padding for alignment)  
  
union Data { int i; float f; char c; };  
// Size = size of largest member
```

3.8 Dynamic Memory Allocation

```
malloc(size)           // allocates, no initialization
calloc(n, size)        // allocates + initializes to 0
realloc(ptr, new_size) // resize
free(ptr)              // deallocate
```

3.9 Common Output Questions – Key Concepts

1. Post-increment (i++) vs Pre-increment (++i)
2. Pointer arithmetic
3. Array indexing with pointers
4. String functions behavior
5. Recursion trace
6. static variable in function (retains value)
7. sizeof for arrays, pointers, structures
8. Bitwise operators: & | ^ ~ << >>
9. Short-circuit evaluation: && and ||
10. Type casting: implicit and explicit

Important Bitwise Tricks:

<code>x & 1</code>	→ check odd/even
<code>x & (x-1)</code>	→ turn off rightmost set bit
<code>x (1 << n)</code>	→ set nth bit
<code>x & ~(1 << n)</code>	→ clear nth bit
<code>x ^ x = 0</code>	→ XOR with itself
<code>x ^ 0 = x</code>	→ XOR with zero



PYQs – C PROGRAMMING

Q1. What is the output?

```
int a = 5;
printf("%d %d %d", a++, ++a, a);
```

Q2. What is the output?

```
char *p = "Hello";
printf("%c", *(p+1));
```

Q3. What is the output?

```
int a[] = {1, 2, 3, 4, 5};  
int *p = a;  
printf("%d", *(p + 3));
```

Q4. What is the output?

```
void fun() {  
    static int x = 0;  
    x++;  
    printf("%d ", x);  
}  
int main() {  
    fun(); fun(); fun();  
    return 0;  
}
```

Q5. What is the output?

```
int x = 10;  
int *p = &x;  
*p = 20;  
printf("%d", x);
```

Q6. What is the output?

```
printf("%d", sizeof("Hello"));
```

Q7. What is the output?

```
int i = 0;  
for(; i < 5; i++);  
printf("%d", i);
```

Q8. What is the difference between `malloc()` and `calloc()` ?

Q9. What is the output?

```
int a = 5, b = 10;
printf("%d", a > b ? a : b);
```

Q10. What is the output?

```
#define SQUARE(x) x*x
printf("%d", SQUARE(3+2));
```

Q11. What is the size of a union that contains an int (4 bytes), a double (8 bytes), and a char (1 byte)?

Q12. What does `strcmp("abc", "abd")` return?

Q13. What is the output?

```
int a = 0;
if(a = 5)
    printf("True");
else
    printf("False");
```

Q14. What is the output?

```
int x = 5;
printf("%d %d", x & 3, x | 3);
```

Q15. How many times is the recursive function called for `fib(5)` using the standard recursive implementation?

OPERATING SYSTEMS

SHORT NOTES

4.1 Process States

New → Ready → Running → Waiting → Ready → Running → Terminated

Process Control Block (PCB): Process ID, state, PC, registers, memory info, I/O info, scheduling info

4.2 CPU Scheduling Algorithms

Algorithm	Preemptive?	Starvation	Convoy Effect
FCFS	No	No	Yes
SJF	No	Yes	No
SRTF (Preemptive SJF)	Yes	Yes	No
Priority	Both possible	Yes	No
Round Robin	Yes	No	No

Key Metrics:

Turnaround Time (TAT) = Completion Time - Arrival Time

Waiting Time (WT) = Turnaround Time - Burst Time

Response Time = First execution time - Arrival Time

Round Robin:

Time quantum too large → becomes FCFS

Time quantum too small → too many context switches

4.3 Process Synchronization

Critical Section Problem – Requirements:

1. **Mutual Exclusion** – Only one process in CS at a time
2. **Progress** – If no one in CS, decision can't be postponed indefinitely
3. **Bounded Waiting** – Limit on number of times other processes enter CS

Semaphores:

```
wait(S) / P(S) / down(S) :  
    while(S <= 0);  
    S--;
```

```
signal(S) / V(S) / up(S) :  
    S++;
```

- **Binary Semaphore (Mutex):** S = 0 or 1
- **Counting Semaphore:** S = any non-negative integer

Classical Problems:

1. Producer-Consumer (Bounded Buffer)
 2. Readers-Writers
 3. Dining Philosophers
-

4.4 Deadlock

Necessary Conditions (ALL 4 must hold):

1. **Mutual Exclusion**
2. **Hold and Wait**
3. **No Preemption**
4. **Circular Wait**

Deadlock Handling:

- Prevention (break one condition)
- Avoidance (Banker's Algorithm)
- Detection & Recovery
- Ignore (Ostrich Algorithm)

Banker's Algorithm:

$Need[i][j] = Max[i][j] - Allocation[i][j]$

Safe State Check:

Work = Available

Find process P_i where $Need[i] \leq Work$

Work = Work + Allocation[i]

Mark P_i as finished

Repeat. If all finished $\rightarrow$ Safe State

Resource Allocation Graph:

Deadlock conditions in RAG:

- Single instance: Cycle $\rightarrow$ Deadlock
 - Multiple instances: Cycle $\rightarrow$ may or may not be deadlock
-

4.5 Memory Management

Fragmentation:

- **Internal:** Allocated memory > Required memory (in fixed partitioning, paging)
- **External:** Total free memory enough but not contiguous (in variable partitioning, segmentation)

Paging: NO external fragmentation, possible internal fragmentation

Segmentation: NO internal fragmentation, possible external fragmentation

Paging:

Logical address = Page Number + Page Offset

Physical address = Frame Number + Page Offset

Page size = Frame size

Number of pages = Virtual address space / Page size

Page table size = Number of pages × Entry size

Page Replacement Algorithms:

Algorithm	Description	Belady's Anomaly?
FIFO	Replace oldest page	Yes
LRU	Replace least recently used	No
Optimal	Replace page not needed for longest time	No
LFU	Replace least frequently used	No

Page Fault Rate:

Effective Access Time = $(1-p) \times ma + p \times (\text{page_fault_time})$

Where p = page fault rate, ma = memory access time

Belady's Anomaly: More frames → more page faults (happens with FIFO)

4.6 Disk Scheduling

Seek Time = Time to move head to desired track

Rotational Latency = Time for desired sector to come under head

Transfer Time = Time to transfer data

Total Access Time = Seek Time + Rotational Latency + Transfer Time

Disk Scheduling Algorithms:

1. **FCFS** – First Come First Served
2. **SSTF** – Shortest Seek Time First (may cause starvation)

3. **SCAN** (Elevator) – Move in one direction, serve all, reverse
 4. **C-SCAN** (Circular SCAN) – One direction only, jump back to start
 5. **LOOK** – Like SCAN but only goes till last request
 6. **C-LOOK** – Like C-SCAN but only goes till last request
-

4.7 File Systems

File Allocation Methods:

Method	Advantages	Disadvantages
Contiguous	Fast access, simple	External fragmentation, file can't grow
Linked	No external fragmentation, file can grow	Sequential access only, pointer overhead
Indexed	Direct access, no external fragmentation	Index block overhead



PYQs – OPERATING SYSTEMS

- Q1.** Four necessary conditions for deadlock are _____.
- Q2.** Given processes with arrival times and burst times, calculate the average waiting time using SJF (non-preemptive).
- Q3.** In a system using paging with page size 4 KB and 32-bit logical address, how many entries are in the page table?
- Q4.** Which page replacement algorithm suffers from Belady's anomaly?
- Q5.** What is the purpose of Banker's algorithm?
- Q6.** In Round Robin scheduling with time quantum = 4, given processes P1(BT=5), P2(BT=3), P3(BT=8), find the average turnaround time.
- Q7.** Which memory management scheme suffers from external fragmentation?
- Q8.** What is the difference between preemptive and non-preemptive scheduling?
- Q9.** A semaphore is initialized to 1. Two processes P1 and P2 use wait(S) and signal(S). What is this called?
- Q10.** In a system with 5 processes and 3 resource types, given the Max, Allocation, and Available matrices, determine if the system is in a safe state.
- Q11.** What is a race condition?

Q12. Which disk scheduling algorithm is most fair and avoids starvation?

Q13. What is the difference between internal and external fragmentation?

Q14. Calculate the number of page faults using LRU for the reference string 7,0,1,2,0,3,0,4,2,3,0,3,2 with 3 frames.

Q15. What is thrashing in operating systems?

COMPUTER NETWORKS

SHORT NOTES

5.1 OSI Model (7 Layers)

Layer 7: Application	– HTTP, FTP, SMTP, DNS, SNMP
Layer 6: Presentation	– Encryption, Compression, Translation
Layer 5: Session	– Session management, synchronization
Layer 4: Transport	– TCP (reliable), UDP (unreliable), Ports
Layer 3: Network	– IP, Routing, Logical addressing, ICMP
Layer 2: Data Link	– MAC addressing, Framing, Error detection
Layer 1: Physical	– Bits, cables, connectors, signals

Mnemonic (top-down): All People Seem To Need Data Processing

5.2 TCP/IP Model (4/5 Layers)

Application (combines OSI 5,6,7)
Transport (TCP/UDP)
Internet/Network (IP)
Network Access/Data Link + Physical

5.3 IP Addressing (IPv4)

Classes:

Class	First Octet	Default Mask	Network/Host
A	1-126	255.0.0.0 (/8)	7 net, 24 host bits

Class	First Octet	Default Mask	Network/Host
B	128–191	255.255.0.0 (/16)	14 net, 16 host bits
C	192–223	255.255.255.0 (/24)	21 net, 8 host bits
D	224–239	Multicast	—
E	240–255	Reserved	—

Special addresses:

- 127.0.0.1 → Loopback
- 0.0.0.0 → Default route
- 255.255.255.255 → Limited broadcast

Subnetting Formulas:

Number of subnets = 2^s (s = borrowed bits)

Number of hosts per subnet = 2^h - 2 (h = remaining host bits)

Subnet mask = default mask + subnet bits set to 1

CIDR notation: /n means n network bits

Block size = 256 - subnet mask octet value

5.4 Important Protocols & Port Numbers

Protocol	Port	Transport	Purpose
HTTP	80	TCP	Web
HTTPS	443	TCP	Secure Web
FTP	20 (data), 21 (control)	TCP	File Transfer
SSH	22	TCP	Secure Shell
Telnet	23	TCP	Remote Login
SMTP	25	TCP	Email Sending
DNS	53	TCP/UDP	Name Resolution
DHCP	67/68	UDP	IP Assignment
POP3	110	TCP	Email Retrieval
IMAP	143	TCP	Email Retrieval
SNMP	161	UDP	Network Mgmt

5.5 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK, retransmission)	Unreliable
Ordering	Ordered delivery	No guarantee
Speed	Slower	Faster
Header size	20 bytes min	8 bytes
Use cases	HTTP, FTP, Email	DNS, Streaming, Gaming

5.6 Error Detection

Parity Bit: Detects single bit errors

Checksum: Sum of segments, complement

CRC (Cyclic Redundancy Check):

Append (divisor length - 1) zeros to data

Divide by generator polynomial (XOR division)

Remainder = CRC bits

Transmitted data = Original data + CRC

Hamming Code:

Number of parity bits r : $2^r \geq m + r + 1$

(m = data bits)

Can detect 2-bit errors and correct 1-bit error

Parity bits at positions 1, 2, 4, 8, 16...

5.7 Transmission Time & Propagation Delay

Transmission Time (T_t) = Packet size / Bandwidth

Propagation Delay (T_p) = Distance / Speed of signal

Total Delay = $T_t + T_p + \text{Queuing delay} + \text{Processing delay}$

Throughput = Data / Total time

Efficiency = $T_t / (T_t + 2T_p)$ [for Stop-and-Wait]

Sliding Window:

Stop and Wait: Window size = 1

Efficiency = $1 / (1 + 2a)$, where $a = T_p/T_t$

Go-Back-N:

Sender window = $2^n - 1$ (n = sequence number bits)

Receiver window = 1

Efficiency = $W / (1 + 2a)$ if $W < 1 + 2a$

Selective Repeat:

Sender window = Receiver window = $2^{(n-1)}$

Efficiency = $W / (1 + 2a)$ if $W < 1 + 2a$

5.8 Network Devices

Device	Layer	Function
Hub	1(Physical)	Broadcasts to all ports
Repeater	1	Amplifies signal
Switch	2(Data Link)	MAC-based forwarding
Bridge	2	Connects two LANs
Router	3(Network)	IP-based routing
Gateway	7(Application)	Protocol conversion

5.9 Routing Algorithms

Distance Vector (Bellman-Ford based): RIP

- Count to infinity problem
- Slow convergence

Link State (Dijkstra based): OSPF

- Each router knows complete topology
 - Faster convergence
-

5.10 MAC Sub-layer Protocols

ALOHA:

Pure ALOHA: Max throughput = $1/(2e) \approx 18.4\%$

Slotted ALOHA: Max throughput = $1/e \approx 36.8\%$

CSMA/CD: Used in Ethernet (IEEE 802.3)

Min frame size = $2 \times T_p \times \text{Bandwidth}$

CSMA/CA: Used in WiFi (IEEE 802.11)



PYQs – COMPUTER NETWORKS

- Q1.** Which layer of OSI model is responsible for routing and logical addressing?
 - Q2.** How many host addresses are available in a /26 subnet?
 - Q3.** What is the minimum number of parity bits required to detect and correct single-bit error in 8-bit data using Hamming code?
 - Q4.** In Go-Back-N protocol with 3-bit sequence number, the maximum sender window size is?
 - Q5.** What is the transmission time for a 1 MB file on a 10 Mbps link?
 - Q6.** Which protocol is used for resolving domain names to IP addresses?
 - Q7.** What is the port number for SMTP?
 - Q8.** The maximum throughput of slotted ALOHA is?
 - Q9.** What is the difference between TCP and UDP?
 - Q10.** In a Class C network, how many usable host addresses are available?
 - Q11.** What is CRC? How is it different from parity check?
 - Q12.** Which layer is responsible for end-to-end error recovery and flow control?
 - Q13.** What is the purpose of ARP protocol?
 - Q14.** Subnet mask 255.255.255.224 corresponds to which CIDR notation?
 - Q15.** Which routing algorithm does OSPF use?
-
-



6 DBMS (Database Management Systems)



SHORT NOTES

6.1 Keys

Key	Definition
Super Key	Any set of attributes that uniquely identifies a tuple
Candidate Key	Minimal super key (no proper subset is a super key)
Primary Key	Chosen candidate key
Foreign Key	Attribute in one table referencing primary key of another
Alternate Key	Candidate keys that are not primary key

6.2 Normal Forms

Normal Form	Condition
1NF	Atomic values (no multivalued attributes), each row unique
2NF	1NF + No partial dependency (on part of composite key)
3NF	2NF + No transitive dependency (non-key $\rightarrow$ non-key)
BCNF	For every FD $X \rightarrow Y$, X must be a superkey

Quick Test:

1NF: No repeating groups

2NF: 1NF + No partial dependency

3NF: 2NF + No transitive dependency

BCNF: Every determinant is a candidate key

Relation: $BCNF \subset 3NF \subset 2NF \subset 1NF$

6.3 Functional Dependencies & Closure

Attribute closure of X (X^+):

Start with X

For each FD $A \rightarrow B$, if $A \subseteq X^+$, add B to X^+

Repeat until no change

If X^+ contains all attributes $\rightarrow X$ is a superkey

If no proper subset of X is a superkey $\rightarrow X$ is a candidate key

Armstrong's Axioms:

1. **Reflexivity:** If $B \subseteq A$, then $A \rightarrow B$
2. **Augmentation:** If $A \rightarrow B$, then $AC \rightarrow BC$
3. **Transitivity:** If $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$

Derived Rules:

4. **Union:** If $A \rightarrow B$ and $A \rightarrow C$, then $A \rightarrow BC$
5. **Decomposition:** If $A \rightarrow BC$, then $A \rightarrow B$ and $A \rightarrow C$
6. **Pseudotransitivity:** If $A \rightarrow B$ and $CB \rightarrow D$, then $AC \rightarrow D$

6.4 Relational Algebra

Operation	Symbol	Description
Selection	σ	Select rows (horizontal)
Projection	π	Select columns (vertical)
Cartesian Product	$\times$	Cross product of two tables
Union	$\cup$	All tuples from both (compatible)
Set Difference	$-$	Tuples in R but not in S
Intersection	$\cap$	Tuples in both R and S
Natural Join	$\bowtie$	Join on common attributes
Rename	ρ	Rename relation/attributes
Division	$\div$	Universal quantifier

Maximum tuples in $R \bowtie S$ (natural join) = $|R| \times |S|$

Minimum tuples = 0

6.5 SQL

```
-- Basic Syntax
SELECT column1, column2 FROM table WHERE condition
GROUP BY column HAVING condition
ORDER BY column ASC/DESC;
```

```

-- Aggregate Functions
COUNT(), SUM(), AVG(), MAX(), MIN()

-- Joins
INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, CROSS JOIN

-- Subqueries
SELECT * FROM emp WHERE sal > (SELECT AVG(sal) FROM emp);

-- Set Operations
UNION, UNION ALL, INTERSECT, EXCEPT/MINUS

-- DDL: CREATE, ALTER, DROP, TRUNCATE
-- DML: SELECT, INSERT, UPDATE, DELETE
-- DCL: GRANT, REVOKE
-- TCL: COMMIT, ROLLBACK, SAVEPOINT

```

Important:

WHERE → before grouping (row-level filter)
HAVING → after grouping (group-level filter)

GROUP BY → used with aggregate functions
ORDER BY → sorting results

NULL comparisons: IS NULL, IS NOT NULL (not = NULL)

6.6 Transactions (ACID Properties)

Property	Meaning
Atomicity	All or nothing
Consistency	DB moves from one consistent state to another
Isolation	Concurrent transactions don't interfere
Durability	Committed changes are permanent

Transaction States:

Active → Partially Committed → Committed
Active → Failed → Aborted

Serializability:

Conflict Serializable: Can create a conflict-equivalent serial schedule

→ Build precedence graph. If NO cycle → Conflict Serializable

View Serializable: $\supseteq$ Conflict Serializable

→ NP-complete to check

Conflict operations: Two operations conflict if:

1. They belong to different transactions
2. They access the same data item
3. At least one is a write

6.7 Concurrency Control

Lock-based:

- **Shared Lock (S):** Read only
- **Exclusive Lock (X):** Read and write

Two-Phase Locking (2PL):

- **Growing phase:** Acquire locks only
- **Shrinking phase:** Release locks only
- Guarantees conflict serializability
- **May have deadlocks**

Timestamp-based: Assigns timestamps to transactions

6.8 ER Model

Entities → Tables

Attributes → Columns

Relationships → Foreign keys or separate tables

Cardinality:

- 1:1 → Foreign key in either table
 - 1:M → Foreign key in M side
 - M:N → Separate relationship table
-



- Q1.** What is the highest normal form of a relation $R(A, B, C, D)$ with FDs: $A \rightarrow B, B \rightarrow C$?
- Q2.** Write a SQL query to find the second highest salary from an Employee table.
- Q3.** What is the difference between WHERE and HAVING clause?
- Q4.** How many candidate keys does a relation $R(A, B, C, D)$ with FDs: $AB \rightarrow C, C \rightarrow D, D \rightarrow A$ have?
- Q5.** What are ACID properties?
- Q6.** What is the result of $\pi_A(\sigma_{\{B > 5\}}(R))$ where R has columns A, B, C?
- Q7.** What is the difference between DELETE, TRUNCATE, and DROP?
- Q8.** In two-phase locking protocol, a transaction that has released a lock can acquire new locks. True or False?
- Q9.** What is the maximum number of tuples in $R \times S$ if R has 5 tuples and S has 3 tuples?
- Q10.** Write a SQL query to find employees who earn more than the average salary.
- Q11.** What is a foreign key?
- Q12.** Which normal form eliminates transitive dependency?
- Q13.** What is a view in SQL?
- Q14.** Given a schedule, determine if it is conflict serializable using precedence graph.
- Q15.** What is the difference between UNION and UNION ALL in SQL?
-
-



NUMBER SYSTEM & BOOLEAN ALGEBRA / DIGITAL LOGIC



SHORT NOTES

7.1 Number System Conversions

Binary (Base 2) $\leftrightarrow$ Decimal (Base 10) $\leftrightarrow$ Octal (Base 8) $\leftrightarrow$ Hexadecimal (Base 16)

Binary to Octal: Group 3 bits from right

Binary to Hex: Group 4 bits from right

Decimal to Binary: Divide by 2, collect remainders (bottom to top)

Complements:

r 's complement of $N = r^n - N$

$(r-1)$'s complement of $N = r^n - 1 - N$

For binary (n digits):

1's complement: Flip all bits

2's complement: 1's complement + 1

7.2 Logic Gates

Gate	Expression	Truth when
AND	$A \cdot B$	Both 1
OR	$A + B$	At least one 1
NOT	A'	Input is 0
NAND	$(A \cdot B)'$	Not both 1
NOR	$(A + B)'$	Both 0
XOR	$A \oplus B$	Odd number of 1s
XNOR	$(A \oplus B)'$	Both same

Universal Gates: NAND, NOR (can implement any gate)

7.3 Boolean Algebra Laws

Identity: $A + 0 = A$, $A \cdot 1 = A$

Null: $A + 1 = 1$, $A \cdot 0 = 0$

Complement: $A + A' = 1$, $A \cdot A' = 0$

Idempotent: $A + A = A$, $A \cdot A = A$

Involution: $(A')' = A$

De Morgan's: $(AB)' = A' + B'$, $(A+B)' = A' \cdot B'$

Absorption: $A + AB = A$, $A(A+B) = A$

Distributive: $A(B+C) = AB+AC$, $A+BC = (A+B)(A+C)$

Consensus: $AB + A'C + BC = AB + A'C$

7.4 Minterms & Maxterms

Minterm (m): Product term where each variable appears once

$SOP = \Sigma(\text{minterms})$ – Canonical Sum of Products

$F = 1$ for listed minterms

Maxterm (M): Sum term where each variable appears once

$POS = \Pi(\text{maxterms})$ – Canonical Product of Sums

$F = 0$ for listed maxterms

Relationship: $SOP \text{ minterms} + POS \text{ maxterms} = \text{all possible terms}$

7.5 K-Map (Karnaugh Map)

2 variables: 4 cells (2×2)

3 variables: 8 cells (2×4)

4 variables: 16 cells (4×4)

Rules:

- 1. Group sizes must be powers of 2 (1, 2, 4, 8, 16)
 - 2. Groups must be rectangular
 - 3. Larger groups → simpler expression
 - 4. Cover all 1s (for SOP) or all 0s (for POS)
 - 5. Don't cares (X) can be used to make larger groups
 - 6. Wraparound allowed
-

7.6 Combinational Circuits

Circuit	Input	Output	Formula
Half Adder	A, B	Sum = $A \oplus B$, Carry = AB	—
Full Adder	A, B, Cin	Sum = $A \oplus B \oplus Cin$, Cout = $AB + BCin + ACin$	—
Half Subtractor	A, B	Diff = $A \oplus B$, Borrow = $A'B$	—
Full Subtractor	A, B, Bin	Diff = $A \oplus B \oplus Bin$, Bout = $A'B + (A \oplus B)'Bin$	—
MUX (2 ⁿ :1)	2 ⁿ inputs + n select	1 output	Can implement any function

Circuit	Input	Output	Formula
DEMUX (1:2 ⁿ)	1 input + n select	2 ⁿ outputs	—
Decoder (n:2 ⁿ)	n inputs	2 ⁿ outputs	Generates all minterms
Encoder (2 ⁿ :n)	2 ⁿ inputs	n outputs	—

MUX as Universal Function Generator:

- 2ⁿ:1 MUX can implement any n-variable function
- (n-1) variables → select lines, nth variable → data inputs (0, 1, x, x')

7.7 Sequential Circuits

Flip-Flops:

Type	Characteristic Equation	Feature
SR	$Q(t+1) = S + R'Q$, $SR = 0$	S=R=1 invalid
JK	$Q(t+1) = JQ' + K'Q$	J=K=1 → toggle
D	$Q(t+1) = D$	Stores data
T	$Q(t+1) = T \oplus Q$	T=1 → toggle

Counters:

n-bit counter counts from 0 to $2^n - 1$

Mod-N counter: Counts N states (0 to N-1)

Needs $\lceil \log_2 N \rceil$ flip-flops



PYQs – NUMBER SYSTEM & BOOLEAN ALGEBRA

Q1. Convert $(1011.101)_2$ to decimal.

Q2. Simplify using K-map: $F(A,B,C,D) = \sum m(0,1,2,5,8,9,10)$

Q3. The 2's complement of 8-bit number 01010110 is?

Q4. How many 2-input NAND gates are required to implement a 2-input OR gate?

Q5. Simplify: $AB + A'C + BC$ using Boolean algebra.

- Q06. An 8:1 MUX needs how many select lines?
- Q07. The output of an XOR gate is 1 when?
- Q08. $(2F.A)_{16} = (?)_{10}$
- Q09. What is the minimum number of flip-flops required for a mod-10 counter?
- Q10. Apply De Morgan's theorem to simplify $(A + B + C)'$.
- Q11. Which flip-flop is known as a delay flip-flop?
- Q12. A 4:16 decoder can generate all minterms of how many variables?
-
-

SOFTWARE ENGINEERING (Brief – Low Weightage)



SHORT NOTES

SDLC Models

Model	Key Feature	When to Use
Waterfall	Sequential, no going back	Requirements well understood
Spiral	Risk-driven, iterative	High risk projects
Iterative/Incremental	Build in parts	Large systems
Agile/Scrum	Adaptive, sprints	Changing requirements
V-Model	Verification & Validation	Each phase has testing
RAD	Rapid prototyping	Time-critical
Prototype	Build prototype first	Requirements unclear

Testing Types

Unit Testing → Integration Testing → System Testing → Acceptance Testing

Black Box: Test functionality without knowing code (equivalence partitioning, boundary value)

White Box: Test internal code structure (statement, branch, path coverage)

Coupling & Cohesion

Coupling (inter-module dependency):

Low coupling = Good

Data < Stamp < Control < External < Common < Content

Cohesion (intra-module relatedness):

High cohesion = Good

Functional > Sequential > Communicational > Procedural > Temporal >

Logical > Coincidental



PYQs – SOFTWARE ENGINEERING

- Q1. Which SDLC model is best suited when requirements are well understood and unlikely to change?
 - Q2. What is the difference between verification and validation?
 - Q3. In which testing methodology, the internal structure of the program is considered?
 - Q4. Which model is risk-driven?
 - Q5. What is the difference between coupling and cohesion? Which is desirable: high coupling or low coupling?
-
-



OVERALL EXAM STRATEGY & HIGH-YIELD TIPS



2025–2026 Trend Analysis

What Changed in NIMCET 2025 (NIT Trichy):

- 1. **Computer Organization** questions became **conceptual and deeper** – not just formula-based

2. Questions on **pipelining hazards, cache design decisions, addressing modes** were more application-oriented
3. **Booth's Algorithm, IEEE 754** representation appeared
4. **Control unit design** (hardwired vs microprogrammed) concepts were asked
5. Code output questions in C remained standard
6. OS and Networks remained at same level

High Priority for 2026

Priority	Topics
 Must Master	Computer Organization (especially pipelining, cache, memory, number representation), Data Structures, C Programming
 Important	Operating Systems, DBMS (SQL), Boolean Algebra
 Good to Know	Computer Networks, Software Engineering



Time Management in Exam

- **Total questions in CS section:** ~20 (out of 120 total)
- **Time per question:** ~2 minutes
- **Marking scheme:** +4 for correct, -1 for wrong
- **Strategy:** Attempt all confident questions first, skip doubtful ones



QUICK REVISION CHECKLIST

- ☐ 2's complement range & IEEE 754 format
- ☐ Pipelining formulas (speedup, throughput, efficiency)
- ☐ Cache memory (mapping, hit rate, AMAT)
- ☐ Addressing modes (at least 8 types)
- ☐ RISC vs CISC differences
- ☐ Booth's algorithm steps
- ☐ Array address calculation (row & column major)
- ☐ BST, AVL, Heap properties
- ☐ Sorting complexities table
- ☐ Graph: BFS, DFS, Dijkstra, MST
- ☐ Stack applications (infix to postfix)

- ☐ C pointers, arrays, recursion, output tracing
- ☐ Process scheduling (FCFS, SJF, RR) calculations
- ☐ Deadlock conditions & Banker's algorithm
- ☐ Page replacement (FIFO, LRU, Optimal)
- ☐ OSI layers & protocols with port numbers
- ☐ Subnetting & IP addressing
- ☐ SQL queries (GROUP BY, HAVING, JOINS)
- ☐ Normal forms (1NF to BCNF)
- ☐ K-map simplification
- ☐ Logic gates & Boolean laws
- ☐ Flip-flop types & characteristic equations

INTERNET & EMAIL – COMPLETE SHORT NOTES FOR NIMCET 2026

New Addition for NIMCET 2026 – This topic is relatively straightforward and scoring. It overlaps with questions asked in **GATE (CS IT), ISRO, SSC, Bank PO, NTA UGC NET, BCA/MCA entrance exams, and various government IT exams.**

PART A: WEB BROWSING – HOW THE INTERNET WORKS

1. WHAT IS THE INTERNET?

Internet = Global network of interconnected networks
using TCP/IP protocol suite

It is a "Network of Networks"

Started as: ARPANET (1969) – US Department of Defense

Key Milestones:

Year	Event
------	-------

Year	Event
1969	ARPANET – first 4 nodes
1971	First email sent (Ray Tomlinson)
1983	TCP/IP adopted – "Birthday of Internet"
1989	WWW invented by Tim Berners-Lee at CERN
1990	First web browser (WorldWideWeb/Nexus)
1991	WWW publicly available
1993	Mosaic browser – first graphical browser

⚠ Internet ≠ WWW

- **Internet** = Physical infrastructure (cables, routers, servers)
- **WWW** = Service/application running ON the internet (collection of web pages)

2. HOW THE INTERNET WORKS – STEP BY STEP

User types URL in browser

↓

Step 1: DNS Resolution

URL → IP Address

(e.g., `www.google.com` → `142.250.182.46`)

↓

Step 2: TCP Connection (3-Way Handshake)

SYN → SYN-ACK → ACK

↓

Step 3: HTTP/HTTPS Request

Browser sends GET/POST request to server

↓

Step 4: Server Processes Request

Web server (Apache, Nginx, IIS) processes

May query database, run server-side code

↓

Step 5: HTTP Response

Server sends back HTML, CSS, JS, images

Status codes: 200 OK, 404 Not Found, etc.

↓

Step 6: Browser Rendering

Browser parses HTML → builds DOM tree

Applies CSS → Renders page

Executes JavaScript

↓

Step 7: TCP Connection Closed (or kept alive)

3. KEY INTERNET TERMINOLOGY

3.1 URL (Uniform Resource Locator)

Structure:

`protocol://domain:port/path/file?query#fragment`

Example:

`https://www.example.com:443/products/page.html?id=5#section2`

Protocol: `https`

Domain: `www.example.com`

Port: `443` (default for HTTPS)

Path: `/products/page.html`

Query: `?id=5`

Fragment: `#section2`

3.2 Domain Name System (DNS)

DNS = "Phone book of the Internet"

Converts domain names → IP addresses

Hierarchy:

Root (.) → TLD (.com, .org, .in) → SLD (google) → Subdomain (www)

DNS Record Types:

A Record → Maps domain to IPv4 address

AAAA Record → Maps domain to IPv6 address

CNAME → Alias for another domain

MX Record → Mail server for the domain

NS Record → Nameserver for the domain

TXT Record → Text info (SPF, verification)

PTR Record → Reverse DNS (IP → domain)

SOA Record → Start of Authority

DNS Resolution Process:

Browser cache → OS cache → Router cache →
ISP DNS (Recursive Resolver) → Root Server →
TLD Server → Authoritative Server → IP returned

DNS Port: 53 (both TCP and UDP)

DNS primarily uses UDP for queries (faster)

Uses TCP for zone transfers and large responses

Top-Level Domains (TLDs):

TLD	Purpose
.com	Commercial
.org	Organizations
.edu	Educational
.gov	Government
.net	Network services
.in	India (Country code)
.mil	Military
.info	Information

3.3 IP Address

IPv4: 32-bit, dotted decimal (e.g., 192.168.1.1)

Total addresses: $2^{32} \approx 4.3$ billion

IPv6: 128-bit, hexadecimal with colons

(e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)

Total addresses: $2^{128} \approx 3.4 \times 10^{38}$

Private IP Ranges (not routable on internet):

Class A: 10.0.0.0 to 10.255.255.255

Class B: 172.16.0.0 to 172.31.255.255

Class C: 192.168.0.0 to 192.168.255.255

Public IP: Assigned by ISP, globally unique

Static IP: Fixed, doesn't change

Dynamic IP: Assigned by DHCP, can change

3.4 Important Internet Protocols

Protocol	Full Form	Port	Purpose
HTTP	HyperText Transfer Protocol	80	Web page transfer (unencrypted)
HTTPS	HTTP Secure (HTTP + SSL/TLS)	443	Encrypted web communication
FTP	File Transfer Protocol	20/21	File upload/download
SFTP	SSH File Transfer Protocol	22	Secure file transfer
SMTP	Simple Mail Transfer Protocol	25/587	Sending emails
POP3	Post Office Protocol v3	110	Download emails (deletes from server)
IMAP	Internet Message Access Protocol	143	Access emails (keeps on server)
DNS	Domain Name System	53	Name resolution
DHCP	Dynamic Host Configuration Protocol	67/68	Auto IP assignment
TELNET	Terminal Network	23	Remote login (unencrypted)
SSH	Secure Shell	22	Secure remote login
SNMP	Simple Network Management Protocol	161	Network monitoring
TCP	Transmission Control Protocol	—	Reliable, connection-oriented
UDP	User Datagram Protocol	—	Unreliable, connectionless, fast
IP	Internet Protocol	—	Addressing and routing
ICMP	Internet Control Message Protocol	—	Ping, error reporting
ARP	Address Resolution Protocol	—	IP → MAC
RARP	Reverse ARP	—	MAC → IP
SSL/TLS	Secure Sockets Layer / Transport Layer Security	—	Encryption

3.5 HTTP Methods & Status Codes

HTTP Methods:

Method	Purpose	Idempotent
GET	Retrieve data	Yes
POST	Submit data	No
PUT	Update/Replace	Yes
DELETE	Delete resource	Yes
PATCH	Partial update	No
HEAD	Like GET but no body	Yes
OPTIONS	Supported methods	Yes

HTTP Status Codes:

Code	Category	Examples
1xx	Informational	100 Continue
2xx	Success	200 OK , 201 Created, 204 No Content
3xx	Redirection	301 Moved Permanently , 302 Found, 304 Not Modified
4xx	Client Error	400 Bad Request , 401 Unauthorized , 403 Forbidden , 404 Not Found , 408 Timeout, 429 Too Many Requests
5xx	Server Error	500 Internal Server Error , 502 Bad Gateway, 503 Service Unavailable

3.6 Web Browsers

Function: Interpret and render HTML, CSS, JavaScript

Popular Browsers:

- Google Chrome (Blink engine)
- Mozilla Firefox (Gecko engine)
- Safari (WebKit engine)
- Microsoft Edge (Blink engine, formerly EdgeHTML)

Browser Components:

1. User Interface (address bar, buttons)
2. Browser Engine (coordinates UI and rendering)
3. Rendering Engine (parses HTML/CSS, renders page)
4. Networking (HTTP requests)
5. JavaScript Interpreter (V8 in Chrome)
6. Data Storage (cookies, cache, localStorage)

Key Browser Concepts:

Concept	Description
Cache	Stores copies of web resources locally for faster loading
Cookies	Small text files stored by websites on user's browser
Session	Server-side storage of user data during a visit
Bookmark	Saved URL for quick access
Plugin/Extension	Add-on software for extra browser functionality
Incognito/Private	No history, cookies, or cache saved locally
Tabs	Multiple pages in same browser window

3.7 Web Technologies

Client-Side (runs on browser):

HTML — Structure/content of web page

CSS — Styling/layout

JavaScript — Interactivity/dynamic behavior

Server-Side (runs on server):

PHP, Python, Java, Node.js, Ruby, ASP.NET

Databases:

MySQL, PostgreSQL, MongoDB, Oracle

Full Stack = Client-Side + Server-Side + Database

Web Architecture:

Static Website: Fixed content (HTML files)

Dynamic Website: Content generated on-the-fly (uses databases)

Client-Server Model:

Client (browser) ↔ Internet ↔ Server (web server + app server + database)

3.8 Search Engines

Function: Index and retrieve web pages based on queries

Popular: Google, Bing, Yahoo, DuckDuckGo, Baidu

How Search Engines Work:

- 1. Crawling – Bots (spiders/crawlers) discover web pages
- 2. Indexing – Store and organize page content
- 3. Ranking – Use algorithms to rank results (e.g., PageRank)

SEO (Search Engine Optimization):

Techniques to improve website visibility in search results

3.9 Types of Internet Connections

Type	Speed	Medium
Dial-up	Up to 56 Kbps	Phone line
DSL	1-100 Mbps	Phone line
Cable	10-500 Mbps	Coaxial cable
Fiber Optic	100 Mbps-10 Gbps	Optical fiber
Wi-Fi	Varies	Radio waves
Satellite	12-100 Mbps	Satellite signal
4G/LTE	10-100 Mbps	Cellular
5G	Up to 10 Gbps	Cellular

3.10 Important Networking Concepts

ISP (Internet Service Provider): Provides internet access

Tier 1 → Tier 2 → Tier 3 → End users

Bandwidth: Maximum data transfer rate (bps)

Latency: Delay in data transmission

Throughput: Actual data transfer rate

Firewall: Filters incoming/outgoing network traffic

Proxy Server: Intermediary between client and server

VPN (Virtual Private Network): Encrypted tunnel for secure communication

NAT (Network Address Translation): Maps private IP to public IP

3.11 Web Page vs Website vs Web Server vs Web Portal

Term	Definition
Web Page	Single HTML document
Website	Collection of related web pages under a domain
Web Server	Software/hardware that serves web pages (Apache, Nginx, IIS)
Web Portal	Gateway providing various services (email, news, search)
Home Page	Main/first page of a website
Web Hosting	Service that stores website files on a server

PART B: EMAIL (Electronic Mail)

4. EMAIL FUNDAMENTALS

4.1 What is Email?

Email = Electronic method of sending/receiving messages over internet

Invented by: Ray Tomlinson (1971) – introduced @ symbol

4.2 Email Address Format

username@domain.tld

Example: john.doe@gmail.com

username: john.doe

@: "at" separator

domain: gmail

TLD: .com

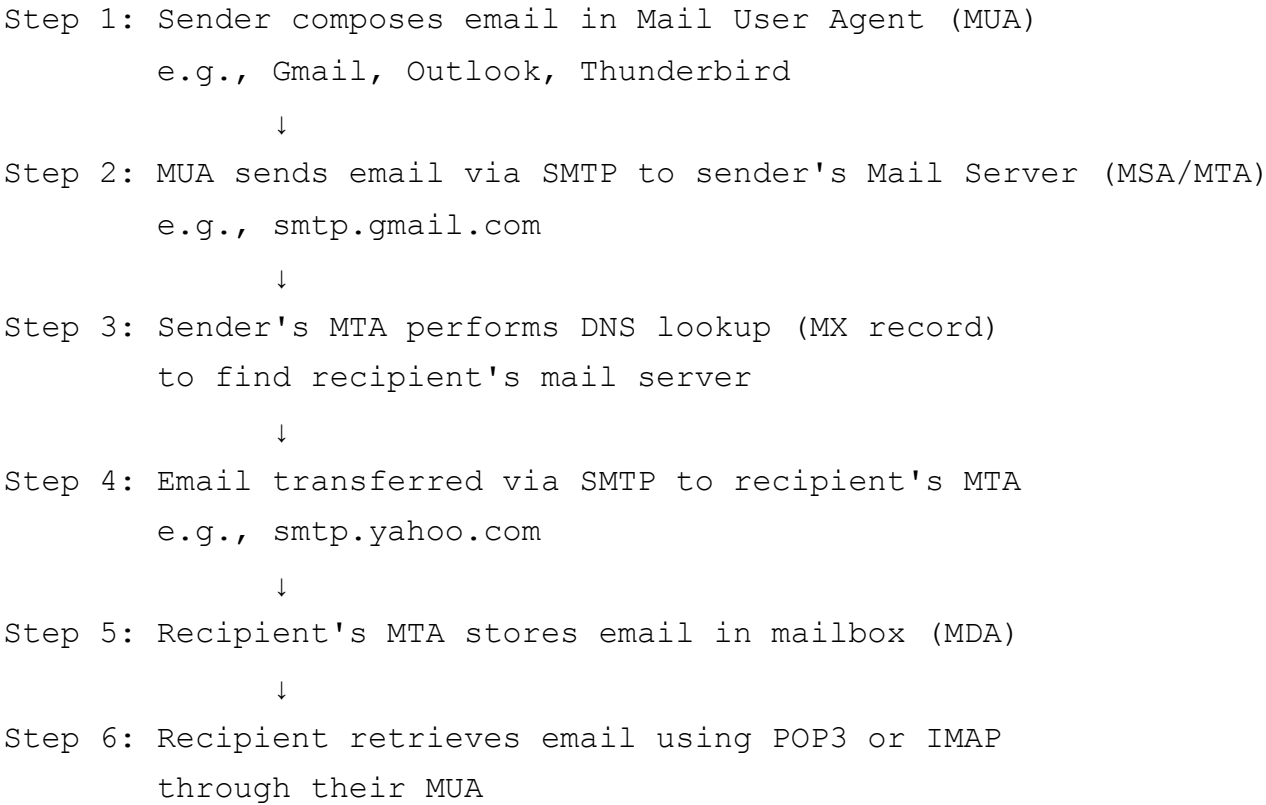
4.3 Email Protocols

Protocol	Port	Function	Direction
SMTP	25 (plain), 587 (TLS), 465 (SSL)	Sending emails	Client → Server, Server → Server
POP3	110 (plain), 995 (SSL)	Receiving (downloads & deletes from server)	Server → Client
IMAP	143 (plain), 993 (SSL)	Receiving (syncs, keeps on server)	Server → Client

SMTP vs POP3 vs IMAP:

Feature	SMTP	POP3	IMAP
Purpose	Send	Receive (download)	Receive (sync)
Server storage	N/A	Deletes after download	Keeps on server
Multi-device	N/A	Difficult	Yes, synced
Offline access	N/A	Yes (downloaded)	Limited
Bandwidth	N/A	Less (downloads once)	More (constant sync)

4.4 How Email Works – Complete Flow



Components:

- MUA (Mail User Agent) → Email client (Gmail app, Outlook)
- MSA (Mail Submission Agent) → Accepts from MUA

- MTA (Mail Transfer Agent) → Routes emails between servers (Sendmail, Postfix)
 - MDA (Mail Delivery Agent) → Delivers to recipient's mailbox
-

4.5 Email Message Structure

HEADER:

From: sender@example.com
To: recipient@example.com
CC: copy@example.com
BCC: blindcopy@example.com
Subject: Email Subject
Date: Timestamp
Message-ID: Unique identifier
Reply-To: Address for replies
Content-Type: text/plain, text/html, multipart/mixed

BODY:

Main content of the email
Can be plain text or HTML formatted

ATTACHMENTS:

Files attached using MIME encoding

4.6 Email Fields Explained

Field	Full Form	Description
To	—	Primary recipient(s)
CC	Carbon Copy	Secondary recipients (visible to all)
BCC	Blind Carbon Copy	Hidden recipients (others can't see)
Subject	—	Brief description of email content
From	—	Sender's email address
Reply-To	—	Address where reply should go
Attachment	—	Files sent with the email

4.7 MIME (Multipurpose Internet Mail Extensions)

Purpose: Extends email to support:

- Non-ASCII text (international characters)
- Attachments (images, documents, audio, video)
- HTML formatted emails
- Multiple parts in one message

MIME Types:

- text/plain → Plain text
- text/html → HTML content
- image/jpeg → JPEG image
- image/png → PNG image
- application/pdf → PDF document
- application/zip → ZIP archive
- audio/mpeg → MP3 audio
- video/mp4 → MP4 video
- multipart/mixed → Email with attachments

Encoding:

- Base64 – Encodes binary data as ASCII text for email transmission
- Quoted-Printable – For mostly ASCII text with some special chars

4.8 Types of Email Services

Type	Description	Examples
Webmail	Accessed through web browser	Gmail, Yahoo Mail, Outlook.com
Desktop Client	Installed software	Microsoft Outlook, Thunderbird, Apple Mail
Mobile App	Mobile email apps	Gmail app, Outlook app

4.9 Email Management Best Practices

1. Organize with Folders/Labels
2. Use Filters/Rules for auto-sorting
3. Archive old emails instead of deleting
4. Unsubscribe from unwanted newsletters
5. Use meaningful subject lines
6. Don't open suspicious attachments
7. Use BCC for mass emails to protect privacy
8. Set up email signatures

- 9. Enable 2-factor authentication
- 10. Regular password changes

4.10 Mailing List & Email Forwarding

Mailing List: Single address that forwards to multiple recipients
e.g., team@company.com → all team members

Email Forwarding: Redirect incoming emails to another address

Auto-Reply/Out of Office: Automatic response when away

Email Alias: Alternative address for same mailbox

PART C: ONLINE SECURITY

5. ONLINE THREATS & SECURITY

5.1 Types of Cyber Threats

5.1.1 Malware (Malicious Software)

Type	Description	Spreads Via
Virus	Attaches to files, replicates when executed	Infected files, email attachments
Worm	Self-replicates WITHOUT host file, spreads via network	Network automatically
Trojan Horse	Disguised as legitimate software	Downloaded by user
Ransomware	Encrypts files, demands payment for decryption	Email, downloads (e.g., WannaCry)
Spyware	Secretly monitors user activity	Bundled software
Adware	Displays unwanted advertisements	Free software
Rootkit	Hides deep in OS, gives attacker root access	Exploits, downloads
Keylogger	Records keystrokes (passwords, credit cards)	Trojans, physical access

Type	Description	Spreads Via
Botnet	Network of infected computers (zombies) controlled remotely	Worms, trojans
Logic Bomb	Triggers malicious action on specific condition/date	Insider threat

Key Difference:

Virus → Needs host file + user action to spread

Worm → No host file needed, self-propagates over network

Trojan → Doesn't replicate, tricks user into installing

5.1.2 Network Attacks

Attack	Description
Phishing	Fake emails/websites to steal credentials
Spear Phishing	Targeted phishing at specific individuals
Whaling	Phishing targeting senior executives
Vishing	Voice phishing (phone calls)
Smishing	SMS phishing
Pharming	Redirects legitimate website traffic to fake site (DNS poisoning)
Man-in-the-Middle (MITM)	Attacker intercepts communication between two parties
Denial of Service (DoS)	Floods server with traffic to make it unavailable
Distributed DoS (DDoS)	DoS from multiple sources (botnet)
SQL Injection	Inserts malicious SQL queries through input fields
Cross-Site Scripting (XSS)	Injects malicious scripts into web pages
Cross-Site Request Forgery (CSRF)	Forces user to execute unwanted actions on authenticated site
Session Hijacking	Steals session ID to impersonate user
IP Spoofing	Faking source IP address
ARP Spoofing	Linking attacker's MAC to legitimate IP
DNS Spoofing/Poisoning	Corrupting DNS cache to redirect traffic
Brute Force Attack	Trying all possible password combinations
Dictionary Attack	Trying common words/passwords

Attack	Description
Rainbow Table Attack	Precomputed hash values for password cracking
Social Engineering	Manipulating people to reveal confidential info
Shoulder Surfing	Observing someone entering password
Dumpster Diving	Searching trash for sensitive information
Eavesdropping/Sniffing	Intercepting network traffic
Replay Attack	Capturing and retransmitting valid data
Zero-Day Attack	Exploiting unknown/unpatched vulnerability

5.2 Security Measures & Countermeasures

5.2.1 Authentication Methods

Something you KNOW → Password, PIN

Something you HAVE → Smart card, OTP device, phone

Something you ARE → Biometrics (fingerprint, face, iris, voice)

Two-Factor Authentication (2FA): Combines two of above

Multi-Factor Authentication (MFA): Two or more factors

Single Sign-On (SSO): One login for multiple services

OAuth: Authorization framework (e.g., "Login with Google")

CAPTCHA: Distinguishes humans from bots

5.2.2 Encryption

Encryption = Converting plaintext to ciphertext using a key

Symmetric Key Encryption:

- Same key for encryption and decryption

- Faster but key distribution problem

- Examples: AES, DES, 3DES, Blowfish, RC4

Asymmetric Key Encryption (Public Key):

- Two keys: Public key (encrypt) + Private key (decrypt)

- Slower but solves key distribution

- Examples: RSA, DSA, ECC, Diffie-Hellman

Hybrid: Use asymmetric to exchange symmetric key, then use symmetric
(This is how HTTPS/SSL/TLS works)

Feature	Symmetric	Asymmetric
Keys	One shared key	Public + Private key pair
Speed	Fast	Slow
Key distribution	Difficult	Easy (public key shared openly)
Examples	AES, DES	RSA, ECC
Use case	Bulk data encryption	Key exchange, digital signatures

5.2.3 Hashing

Hash Function: One-way function that produces fixed-size output
(hash/digest)

Cannot reverse hash to get original input

Used for: Password storage, data integrity, digital signatures

Properties:

1. Deterministic – Same input → same output
2. Fixed output size
3. One-way (pre-image resistance)
4. Collision resistant – Hard to find two inputs with same hash
5. Avalanche effect – Small input change → very different hash

Common Hash Algorithms:

MD5 → 128-bit output (BROKEN – not secure)

SHA-1 → 160-bit output (BROKEN – deprecated)

SHA-256 → 256-bit output (secure, used in Bitcoin)

SHA-512 → 512-bit output

bcrypt → Password hashing (includes salt)

5.2.4 Digital Signatures & Certificates

Digital Signature:

Sender hashes message → Encrypts hash with PRIVATE key

Receiver decrypts hash with sender's PUBLIC key → Verifies

Provides: Authentication + Integrity + Non-repudiation

Digital Certificate:

Issued by Certificate Authority (CA)

Contains: Owner's public key + Owner info + CA's digital signature

Standard: X.509

Examples of CAs: VeriSign, DigiCert, Let's Encrypt

PKI (Public Key Infrastructure):

Framework for managing digital certificates and public keys

5.2.5 SSL/TLS & HTTPS

SSL (Secure Sockets Layer) → Older, deprecated

TLS (Transport Layer Security) → Current standard (TLS 1.2, 1.3)

HTTPS = HTTP + SSL/TLS

Port 443

Provides: Encryption + Authentication + Data Integrity

Indicated by:  padlock icon in browser

SSL/TLS Handshake (simplified):

1. Client Hello (supported ciphers, TLS version)
 2. Server Hello (chosen cipher, server certificate)
 3. Client verifies certificate with CA
 4. Key exchange (generate session key)
 5. Encrypted communication begins
-

5.2.6 Firewalls

Firewall = Network security device that monitors and filters traffic

Types:

1. Packet Filter – Checks headers (IP, port, protocol)
2. Stateful Inspection – Tracks connection state
3. Application Layer (Proxy) – Inspects content at application level
4. Next-Gen Firewall (NGFW) – Deep packet inspection + IPS

Can be:

Hardware firewall (dedicated device)

Software firewall (Windows Defender Firewall, iptables)

5.2.7 Other Security Tools & Concepts

Tool/Concept	Description
Antivirus	Detects, prevents, removes malware
Anti-spyware	Specifically targets spyware
IDS (Intrusion Detection System)	Detects suspicious activity (alerts only)
IPS (Intrusion Prevention System)	Detects AND blocks suspicious activity
VPN	Encrypted tunnel for secure remote access
Proxy Server	Intermediary that can filter content, cache, anonymize
DMZ (Demilitarized Zone)	Network segment between internal and external networks
Honeypot	Decoy system to attract and study attackers
Sandbox	Isolated environment to test suspicious programs
Patch Management	Regular software updates to fix vulnerabilities
Penetration Testing	Authorized simulated attack to find vulnerabilities
Ethical Hacking	Legal hacking to improve security

5.2.8 Password Security Best Practices

- ✓ Use strong passwords (12+ chars, mix of upper, lower, numbers, symbols)
- ✓ Use different passwords for different accounts
- ✓ Enable Two-Factor Authentication (2FA)
- ✓ Use password manager (LastPass, 1Password, Bitwarden)
- ✓ Change passwords regularly
- ✓ Don't share passwords
- ✗ Don't use personal info (name, DOB) in passwords
- ✗ Don't write passwords on paper/sticky notes
- ✗ Don't use common passwords (123456, password, qwerty)

5.2.9 Safe Browsing Practices

1. Check for HTTPS (🔒) before entering sensitive info
2. Don't click suspicious links in emails
3. Verify website URLs carefully (watch for typosquatting)
4. Keep browser and OS updated
5. Use ad-blockers and anti-tracking extensions
6. Don't download from untrusted sources
7. Clear browsing history and cookies regularly
8. Use private/incognito mode on public computers

9. Be cautious with public Wi-Fi (use VPN)
 10. Don't save passwords in public/shared computers
 11. Log out from accounts after use
 12. Verify email sender before responding
-

5.2.10 Email Security

Threats:

- Phishing emails
- Malicious attachments
- Email spoofing
- Spam

Protection:

- SPF (Sender Policy Framework): Verifies sender's IP
- DKIM (DomainKeys Identified Mail): Digital signature for emails
- DMARC: Combines SPF + DKIM for email authentication
- Spam filters
- Don't open attachments from unknown senders
- Don't click links in suspicious emails
- Verify sender's email address carefully
- Use email encryption (PGP/GPG, S/MIME)

PGP (Pretty Good Privacy):

End-to-end encryption for emails
Uses hybrid encryption (RSA + AES)

S/MIME (Secure/MIME):

Email encryption and digital signing standard

5.3 Network Security Concepts

CIA Triad (Core Security Principles):

- C – Confidentiality (only authorized access)
- I – Integrity (data not tampered)
- A – Availability (system accessible when needed)

Additional:

Authentication – Verify identity
Authorization – What you're allowed to do

Non-repudiation – Cannot deny an action

Accountability – Actions can be traced to user

5.4 Cookies & Privacy

Cookies:

First-party cookies: Set by website you visit (login, preferences)

Third-party cookies: Set by other domains (tracking, ads)

Session Cookie: Temporary, deleted when browser closes

Persistent Cookie: Stored for specified duration

Secure Cookie: Only sent over HTTPS

HttpOnly Cookie: Not accessible by JavaScript (prevents XSS)

Privacy Concerns:

- Tracking across websites
- Building user profiles
- Targeted advertising

GDPR (General Data Protection Regulation):

EU regulation for data privacy

Requires consent for data collection

Right to be forgotten

IT Act 2000 (India):

Legal framework for electronic governance

Addresses cyber crimes

Section 43: Unauthorized access penalty

Section 66: Computer-related offences

Section 67: Publishing obscene content

5.5 Important Cyber Security Terms – Quick Reference

Term	Meaning
Vulnerability	Weakness in system that can be exploited
Exploit	Code/technique that takes advantage of vulnerability
Threat	Potential cause of unwanted incident
Risk	Probability of threat exploiting vulnerability

Term	Meaning
Patch	Software update to fix vulnerability
Backdoor	Hidden entry point bypassing normal authentication
Spoofing	Pretending to be someone/something else
Sniffing	Capturing network packets
Encryption	Converting data to unreadable form
Decryption	Converting ciphertext back to plaintext
Plaintext	Original readable data
Ciphertext	Encrypted unreadable data
Cipher	Algorithm for encryption/decryption
Key	Secret value used in encryption



STANDARD QUESTIONS – FROM COMPETITIVE EXAMS

These questions cover the types asked in **GATE, ISRO, UGC NET, SSC, Bank IT Officer, BCA/MCA Entrance**, and predicted for **NIMCET 2026**.

SECTION A: WEB BROWSING & INTERNET (30 Questions)

Q1. Who invented the World Wide Web (WWW)?

(a) Vint Cerf (b) Tim Berners-Lee (c) Bill Gates (d) Ray Tomlinson

Q2. What does URL stand for?

Q3. Which protocol is used to securely transfer data between a web browser and a web server?

(a) HTTP (b) FTP (c) HTTPS (d) SMTP

Q4. Which of the following is NOT a web browser?

(a) Chrome (b) Firefox (c) Apache (d) Safari

Q5. The default port number for HTTP is:

(a) 21 (b) 25 (c) 80 (d) 443

Q06. DNS translates:

- (a) IP address to MAC address
- (b) Domain name to IP address
- (c) MAC address to IP address
- (d) Domain name to MAC address

Q07. Which DNS record type is used for email routing?

- (a) A record (b) CNAME record (c) MX record (d) PTR record

Q08. What does ISP stand for?

Q09. Which of the following is a search engine?

- (a) Linux (b) Google (c) Python (d) Oracle

Q10. The correct URL format is:

- (a) domain://protocol/path
- (b) protocol://domain/path
- (c) path://protocol/domain
- (d) domain/protocol://path

Q11. HTTP status code 404 represents:

- (a) OK (b) Not Found (c) Internal Server Error (d) Forbidden

Q12. Which HTTP method is used to retrieve data from a server?

- (a) POST (b) PUT (c) GET (d) DELETE

Q13. What is the full form of TCP/IP?

Q14. IPv4 address consists of how many bits?

- (a) 16 (b) 32 (c) 64 (d) 128

Q15. IPv6 address consists of how many bits?

- (a) 32 (b) 64 (c) 128 (d) 256

Q16. Which of the following is a private IP address?

- (a) 172.16.5.4 (b) 8.8.8.8 (c) 200.1.1.1 (d) 151.10.5.1

Q17. What is the purpose of a proxy server?

Q18. Which layer of the OSI model does HTTP operate at?

- (a) Transport (b) Network (c) Session (d) Application

Q19. ARPANET was the precursor to:

- (a) WWW (b) Internet (c) Ethernet (d) Wi-Fi

Q20. Which of the following is NOT a top-level domain?

- (a) .com (b) .html (c) .org (d) .edu

Q21. A cookie in the context of web browsing is:

- (a) A virus (b) A small data file stored by websites on user's browser
- (c) A type of firewall (d) An encryption algorithm

Q22. What is the difference between the Internet and the World Wide Web?

Q23. What is web hosting?

Q24. Which protocol is used for automatic IP address assignment?

- (a) DNS (b) DHCP (c) ARP (d) HTTP

Q25. Incognito/Private browsing mode ensures:

- (a) Complete anonymity on internet
- (b) No browsing history saved locally
- (c) Encryption of all data
- (d) Protection from all malware

Q26. What is a web portal? Give an example.

Q27. Which technology is used for creating the structure of a web page?

- (a) CSS (b) JavaScript (c) HTML (d) PHP

Q28. The process by which a search engine discovers web pages is called:

- (a) Indexing (b) Crawling (c) Ranking (d) Caching

Q29. What does HTTPS use for encryption?

- (a) AES (b) SSL/TLS (c) DES (d) MD5

Q30. What is the function of a web server? Name two popular web servers.

SECTION B: EMAIL (20 Questions)

Q31. Which protocol is used for sending emails?

- (a) POP3 (b) IMAP (c) SMTP (d) FTP

Q32. Which protocol downloads emails from server and typically deletes them from the server?

- (a) SMTP (b) POP3 (c) IMAP (d) HTTP

Q33. The default port number for SMTP is:

- (a) 21 (b) 25 (c) 80 (d) 110

Q34. What does BCC stand for in email?

Q35. Who invented email and introduced the @ symbol?

- (a) Tim Berners-Lee (b) Ray Tomlinson (c) Vint Cerf (d) Dennis Ritchie

Q36. What is the difference between POP3 and IMAP?

Q37. MIME stands for:

- (a) Multiple Internet Mail Extensions
- (b) Multipurpose Internet Mail Extensions
- (c) Multiple Internal Mail Encryption
- (d) Multipurpose Internal Mail Extensions

Q38. Which of the following is NOT a part of an email address?

- (a) Username (b) @ symbol (c) Domain name (d) Protocol name

Q39. The port number for POP3 is:

- (a) 25 (b) 80 (c) 110 (d) 143

Q40. The port number for IMAP is:

- (a) 25 (b) 110 (c) 143 (d) 443

Q41. What is email spoofing?

Q42. Which encoding is used by MIME to transmit binary files via email?

- (a) ASCII (b) Base64 (c) Unicode (d) EBCDIC

Q43. What is the purpose of CC in email?

Q44. Which component in email architecture is responsible for transferring emails between mail servers?

- (a) MUA (b) MTA (c) MDA (d) MSA

Q45. What is the difference between webmail and desktop email client?

Q46. SPF, DKIM, and DMARC are related to:

- (a) Email encryption (b) Email authentication (c) Email formatting (d) Email compression

Q47. PGP (Pretty Good Privacy) is used for:

- (a) Web browsing (b) Email encryption (c) File transfer (d) Database management

Q48. What is spam in the context of email?

Q49. Which of the following is a valid email address format?

- (a) user@domain.com (b) user@.com (c) @domain.com (d) user.domain.com

Q50. An auto-reply or out-of-office message is an example of:

- (a) Email forwarding (b) Email filtering (c) Automatic response feature (d) Email encryption

SECTION C: ONLINE SECURITY (30 Questions)

Q51. Which of the following is a self-replicating malware that doesn't need a host file?

- (a) Virus (b) Worm (c) Trojan (d) Adware

Q52. Phishing is:

- (a) A type of virus
- (b) Fraudulent attempt to steal sensitive information through fake communications
- (c) A firewall technique
- (d) An encryption algorithm

Q53. What does the CIA triad stand for in cybersecurity?

Q54. Which type of attack floods a server with excessive requests to make it unavailable?

- (a) Phishing (b) SQL Injection (c) DoS/DDoS (d) Man-in-the-Middle

Q55. A firewall is used to:

- (a) Detect viruses (b) Encrypt data (c) Monitor and filter network traffic (d) Speed up internet

Q56. What is the difference between symmetric and asymmetric encryption?

Q57. RSA is an example of:

- (a) Symmetric encryption (b) Asymmetric encryption (c) Hashing algorithm (d) Firewall

Q58. AES is an example of:

- (a) Symmetric encryption (b) Asymmetric encryption (c) Hashing algorithm (d) Protocol

Q59. SHA-256 is an example of:

- (a) Encryption algorithm (b) Hashing algorithm (c) Protocol (d) Firewall

Q60. What is a digital signature and what does it provide?

Q61. Two-factor authentication (2FA) combines:

- (a) Two passwords
- (b) Two different authentication factors
- (c) Two firewalls
- (d) Two antivirus programs

Q62. A Trojan horse is different from a virus because:

- (a) It self-replicates
- (b) It disguises as legitimate software and doesn't self-replicate
- (c) It only affects servers
- (d) It is a hardware device

Q63. What is ransomware? Give an example.

Q64. What is SQL injection?

Q65. Which of the following is the most secure password?

(a) password123 (b) John1990 (c) P@\$\$w0rd (d) xK9#mQ2\$pL7!

Q66. What is a VPN and why is it used?

Q67. Which protocol provides secure remote login?

(a) Telnet (b) FTP (c) SSH (d) HTTP

Q68. What is the purpose of a digital certificate?

Q69. Which organization issues digital certificates?

(a) ISP (b) Certificate Authority (CA) (c) DNS Server (d) Proxy Server

Q70. What is social engineering in the context of cybersecurity?

Q71. What is the difference between IDS and IPS?

Q72. A honeypot in cybersecurity is:

(a) A type of malware

(b) A decoy system designed to attract attackers

(c) An encryption tool

(d) A type of firewall

Q73. What is a zero-day vulnerability?

Q74. HTTPS uses which default port?

(a) 80 (b) 21 (c) 443 (d) 8080

Q75. What is the difference between a virus and a worm?

Q76. Which of the following is NOT a biometric authentication method?

(a) Fingerprint (b) Iris scan (c) Password (d) Face recognition

Q77. What is cross-site scripting (XSS)?

Q78. What is the purpose of CAPTCHA?

Q79. Which encryption algorithm was used as the standard before AES?

(a) RSA (b) DES (c) MD5 (d) SHA

Q80. What is the IT Act 2000 in India related to?

SECTION D: MIXED/APPLICATION-BASED QUESTIONS (10 Questions)

Q81. Match the following:

- | | |
|-----------|---------------------|
| (i) SMTP | (a) Web browsing |
| (ii) HTTP | (b) Sending email |
| (iii) FTP | (c) Name resolution |
| (iv) DNS | (d) File transfer |

Q82. Arrange the following in the correct order of email delivery:

MUA → MTA → MDA → Recipient's MUA

Q83. A user notices a padlock icon (🔒) in the browser's address bar. This indicates:

- (a) The website is virus-free
- (b) The connection is encrypted using SSL/TLS
- (c) The website is a government site
- (d) Ad-blocker is active

Q84. Which of the following is NOT a function of a web browser?

- (a) Rendering HTML pages
- (b) Executing JavaScript
- (c) Storing databases
- (d) Managing cookies

Q85. What is the output if a user sends an email to an address with an invalid domain?

- (a) Email is silently discarded
- (b) Bounce-back / Non-Delivery Report (NDR) is received
- (c) Email reaches spam folder
- (d) ISP blocks the sender

Q86. A company wants to ensure that emails received are genuinely from the claimed sender domain. Which technology should they implement?

- (a) SSL
- (b) VPN
- (c) SPF/DKIM/DMARC
- (d) Firewall

Q87. Which of the following attacks can be prevented by using HTTPS instead of HTTP?

- (a) SQL Injection
- (b) Man-in-the-Middle
- (c) DDoS
- (d) Social Engineering

Q88. In the URL <https://mail.google.com:443/inbox?user=john#compose>, identify:

- Protocol
- Subdomain
- Domain
- Port
- Path
- Query parameter
- Fragment

Q89. Which of the following stores data on the client side?

- (a) Session
- (b) Cookie
- (c) Database
- (d) Server cache



EXPECTED QUESTION DISTRIBUTION FOR NIMCET 2026

Based on the syllabus description: "Internet and Email: Web Browsing, Email, Online Security"

Sub-topic	Expected Questions	Difficulty
Internet Basics & Protocols	1-2	Easy
Email Protocols & Working	1-2	Easy-Medium
Online Security & Threats	1-2	Easy-Medium
Total from this topic	3-5	Mostly Easy



2025 TREND-BASED PREDICTIONS FOR 2026

What to expect:

- ✔ Direct factual questions (port numbers, protocol names)
- ✔ Difference-based questions (POP3 vs IMAP, HTTP vs HTTPS)
- ✔ Matching type (protocol ↔ port ↔ function)
- ✔ Scenario-based security questions
- ✔ URL parsing / email address format
- ✔ Identify the attack type from description
- ✔ Encryption basics (symmetric vs asymmetric)
- ⚠ May include deeper questions following 2025 COA trend

HIGH-YIELD Facts to Memorize:

- SMTP = 25 (send), POP3 = 110 (receive-delete), IMAP = 143 (receive-sync)
- HTTP = 80, HTTPS = 443, FTP = 20/21, SSH = 22, DNS = 53
- Tim Berners-Lee = WWW, Ray Tomlinson = Email (@)
- ARPANET = 1969, WWW = 1989, TCP/IP = 1983
- Virus needs host, Worm self-replicates, Trojan doesn't replicate

- Symmetric (AES, DES) = same key, Asymmetric (RSA) = public+private
 - CIA = Confidentiality + Integrity + Availability
 - Digital Signature = Hash + Private Key encryption
 - HTTPS = HTTP + SSL/TLS = Port 443
 - Firewall = filters traffic, VPN = encrypted tunnel
 - Phishing = fake emails, Pharming = DNS poisoning
 - IPv4 = 32-bit, IPv6 = 128-bit
 - CC = visible copy, BCC = hidden copy
 - MIME = enables attachments in email, Base64 encoding
 - MUA → MTA → MDA (email flow)
-



QUICK REVISION TABLE – PROTOCOLS & PORTS

Protocol	Port	Purpose
FTP	20/21	File Transfer
SSH	22	Secure Remote Login
Telnet	23	Remote Login (plain)
SMTP	25	Send Email
DNS	53	Name Resolution
DHCP	67/68	IP Assignment
HTTP	80	Web (unencrypted)
POP3	110	Receive Email (delete)
IMAP	143	Receive Email (sync)
HTTPS	443	Web (encrypted)
SMTPS	587	Send Email (TLS)
POP3S	995	POP3 (SSL)
IMAPS	993	IMAP (SSL)



Pro Tip: This topic is the **easiest scoring area** in NIMCET CS section. Memorize the key facts, protocols, port numbers, and security concepts. Most questions will be **direct and factual**. Don't overthink – just recall and answer! 🎯